what happens inside the singularities. For that, we need the quantum gravity laws.

In 1962 I moved from Caltech (my undergraduate school) to Princeton University, to study for a PhD in physics. I chose Princeton because John Wheeler taught there. Wheeler was that era's most creative genius, when it comes to Einstein's relativistic laws. I wanted to learn from him.

One September day, with trepidation I knocked on the door of Professor Wheeler's office. It would be my first meeting with the great man. He greeted me with a warm smile, ushered me in, and immediately —as though I were an esteemed colleague, not a total novice—began discussing the mysteries of the implosions of stars. Implosions that produce black holes with singularities in their cores. These singularities, he asserted, "are a place in which the fiery marriage of Einstein's relativistic laws with the quantum laws is consummated." The fruits of that marriage, the laws of quantum gravity, come into full blossom in singularities, Wheeler asserted. If we could understand singularities, we would learn the laws of quantum gravity. Singularities are a rosetta stone for deciphering quantum gravity.